FINAL YEAR PROJECT — 2026

A.U.R.A.

Air-gapped Unified Responsive Agent

A locally-run multi-agent system designed for secure computer & OS automation
— running entirely offline on your machine.



[TEAM MEMBERS]

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// AIR-GAPPED · PRIVATE · LOCAL-FIRST

SECTION 01 — WHO WE ARE

Built by three engineers.

LEAD · SYSTEMS

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LEAD · AGENTS

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LEAD · SECURITY

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SECTION 02 — PROJECT SCOPE

Four domains. One unified agent.

A.U.R.A. crosses the boundaries of artificial intelligence, computer automation, cybersecurity and edge computing to deliver a single safe local agent.

01 | Artificial Intelligence

Local LLMs · Multi-Agent Architecture · Task Planning & Decomposition · Local Reasoning

02 | Computer Automation

OS & App Interaction · Accessibility-Based Automation · Background Tasks · Vision-Based UI Fallback

03 | Cybersecurity

Zero-Trust Execution · Prompt-Injection Defense · Restricted Permissions · Human Approval for High-Risk Actions

04 | Edge Computing

Offline Inference · Quantized Models · Raspberry Pi Companion Node · Encrypted Local Data

THE PROBLEM

AI can reason.
Can we safely let it act on a computer?

Modern AI systems can understand instructions and perform increasingly complex tasks — but giving an AI access to a real computer introduces five critical challenges.

01 Privacy

Sensitive files, credentials, browser sessions and user context may leak outside the local environment.

02 Security

External webpages and documents can contain malicious instructions capable of influencing tool-using AI agents.

03 System Access

An AI assistant needs controlled access to applications and OS resources to perform real tasks.

04 Reliability

Traditional automation fails when application interfaces or layouts change.

05 User Control

High-impact actions should not execute without meaningful human authorization.

[THE GAP]

There is no off-the-shelf assistant that combines **local intelligence**, **real OS access**, and **physical user oversight** — without sending your data to the cloud.

PROPOSED SOLUTION

[A LOCALLY-RUN MULTI-AGENT SYSTEM]

A.U.R.A.

Air-gapped Unified Responsive Agent

A **locally-run multi-agent system** designed to perform computer and OS tasks through controlled execution.

[THE FIVE PILLARS]

Local Intelligence
Quantized language models perform reasoning locally.

Multi-Agent Orchestration
Brain → Manager → Specialized Agent architecture.

Resilient Automation
Accessibility APIs + vision fallback for any UI.

Security by Design
Sanitized inputs, restricted execution permissions.

Human Oversight
High-risk actions paused for biometric approval.

[HOW A.U.R.A. WORKS → NEXT SLIDE]

From a natural-language command to a verified computer action — in seven steps.

HOW A.U.R.A. WORKS

From natural language to controlled action.

[7-STAGE PIPELINE]

01
USER COMMAND
Natural Language
User instruction arrives via desktop or IoT.

02
BRAIN
Intent
Interprets the user's intent.

03
MANAGER
Planning
Decomposes task and selects agent.

04
AGENT
Specialized
Browser, File/OS, Research, Comms, Sec.

05
EXECUTE
Accessibility
Direct UI interaction via structured APIs.

06
FALLBACK
Vision
Local vision model locates UI element.

07
VERIFY
Result
Confirms result, updates relevant state.

[SPECIALIZED AGENTS]
Browser · File / OS · Research · Communication · Security

// ONE NATURAL-LANGUAGE COMMAND → SEVEN CONTROLLED STEPS

KEY FEATURES

Nine capabilities.
A single coherent agent.

[CORE CAPABILITIES]
09

01 | ORCHESTRATION

Multi-Agent Orchestration

Brain, Manager and specialized agents coordinate complex tasks.

02 | REASONING

Hybrid Local Reasoning

BitNet 2B for fast routing. Quantized 7–8B for complex reasoning.

03 | AUTOMATION

Background OS Automation

Accessibility APIs target application elements directly.

04 | RESILIENCE

Vision Fallback

Local vision-based UI detection when accessibility is unavailable.

05 | MEMORY

Long-Term Memory

Stores user preferences, habits and file references across sessions.

06 | SECURITY

Zero-Trust Security

External content sanitized. Execution permissions restricted.

07 | HITL

Biometric HITL Approval

High-risk actions paused until fingerprint verification.

08 | EDGE

Raspberry Pi Companion

Wake-word mic, fingerprint sensor, local visual feedback.

09 | STORAGE

Encrypted Local Storage

Memory and state databases protected with AES-256.

HYBRID LOCAL REASONING

Two-tier local AI.

Right model for the right task.

[ARCHITECTURE]
2B + 7–8B

LIGHTWEIGHT REASONING

BitNet

b1.58 · 2B params

- Fast routing
- Simple decisions
- Classification
- Low-resource inference

COMPLEX REASONING

Quantized

7–8B params

- Task planning
- Context interpretation
- Multi-step reasoning
- Complex decisions

[WHY HYBRID?]

Simple task → Lightweight model

Complex task → Larger model

Goal: use local resources efficiently while keeping inference offline.

Accessibility first.
Vision when necessary.

[APPROACH]
A → V

PRIMARY METHOD

Accessibility APIs

- Identify application elements
- Target UI components directly
- Reduce dependence on screen coordinates

[DECISION]

Element found?

YES

NO

FALLBACK METHOD

Vision Fallback

01 Screenshot

02 Local Vision Model

03 Locate UI Element

04 Continue Task

[RESULT]

A.U.R.A. uses **structured UI information** when available, and **visual understanding** when it is not.

SECURITY & PRIVACY

Security built into the architecture.

[DEFENSE LAYERS]
05

01 | **Local Processing**
AI reasoning and relevant user context remain locally controlled.

02 | **Untrusted Input Handling**
External web content is treated as untrusted and sanitized before reaching the AI.

03 | **Restricted Execution**
Agents operate with controlled permissions rather than unrestricted system access.

04 | **High-Risk Action Control**
High-impact actions are paused until physical biometric approval is provided.

05 | **Encrypted Storage**
Local memory and state are protected using AES-256 encryption.



[INPUT]
SANITIZED
EXTERNAL DATA

[EXECUTION]
RESTRICTED
PERMISSIONS

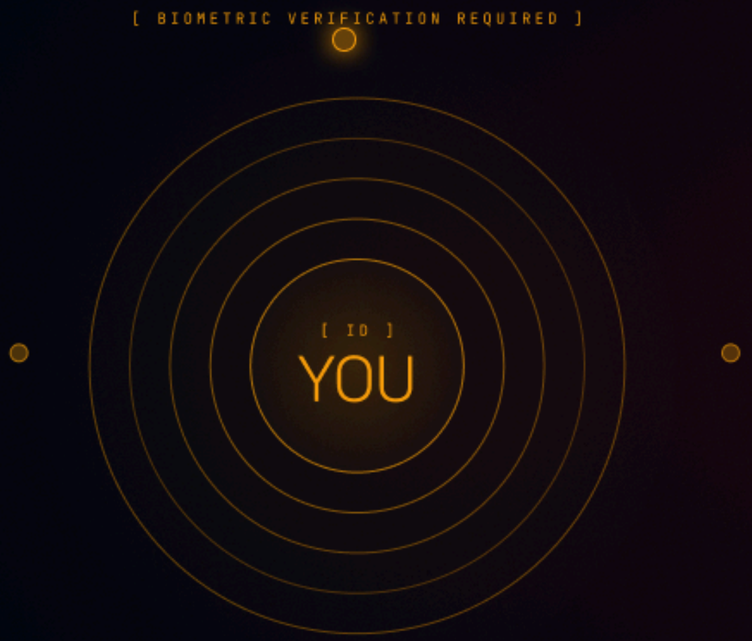
[OVERSIGHT]
BIOMETRIC
CONFIRMATION

[STORAGE]
LOCAL
ENCRYPTED VAULT

HIGH-RISK ACTION CONTROL

Six steps before any irreversible action.

[APPROVAL GATE]
06 STEPS



A.U.R.A. asks for physical confirmation before it executes anything irreversible.

01 **Action Requested**
A specialized agent triggers a high-impact operation.

02 **Risk Check**
The action is classified against A.U.R.A.'s risk policy.

03 **Action Frozen**
Execution is suspended until authorization is secured.

04 **Fingerprint Verification**
Raspberry Pi companion requests a live biometric scan.

05 **User Approval**
The user explicitly confirms intent on-device.

06 **Execution**
Only now does A.U.R.A. perform the action — and only within permitted scope.

SYSTEM ARCHITECTURE

A.U.R.A. from the inside.

[LAYERS]
05

[L1] **User Interaction**	DESKTOP INTERFACE → Natural Language · Task Status · Execution Feedback		IOT COMPANION (NEXT SLIDE →) → Raspberry Pi · Wake Word · Fingerprint Sensor · Local Feedback
[L2] **Orchestration**	BRAIN Intent interpretation	MANAGER Planning + Routing + Shared context	SPECIALIZED AGENTS Browser · File / OS · Research · Communication · Security
[L3] **Intelligence**	BITNET B1.58 · 2B Lightweight reasoning	QUANTIZED 7-8B Complex reasoning	LOCAL VISION MODEL UI fallback
[L4] **Execution**	ACCESSIBILITY APIS Direct UI interaction		RESTRICTED EXECUTION ENV Controlled permissions
[L5] **Data**	CHROMADB Long-term memory	SQLITE Local state	AES-256 Data protection

EDGE COMPANION

A pocket-sized sovereign node.

[HARDWARE]
RPi

WAKE-WORD MICROPHONE

Always-on local listening.

Audio is processed entirely on-device. No audio ever leaves the user's hardware.

FINGERPRINT SENSOR

Physical human confirmation.

High-risk operations are gated by a live biometric scan from the companion device.

LOCAL VISUAL FEEDBACK

A glance says it all.

A discreet LED ring communicates A.U.R.A.'s state — listening, thinking, acting, awaiting approval.

ENCRYPTED LOCAL DATA

Nothing leaves this node.

All companion-side data — audio, biometrics, sessions — stays encrypted on the device.

RASPBERRY PI COMPANION

The hardware that anchors A.U.R.A.

FORM FACTOR
Compact · always-on

INPUTS
Wake-word mic · FP sensor

OUTPUT
LED feedback ring

STORAGE
AES-256 encrypted

[WHY IT MATTERS]

The companion node is an **independent verifier**. It proves that a real human — not a remote attacker — approved every high-risk action.

CONCLUSION

A.U.R.A. turns
natural language into verified action.

[RECAP]

04 PILLARS

A.U.R.A.

Air-gapped Unified Responsive Agent

A **local AI system** designed to turn natural-language commands into **controlled computer actions**.

[01]

Local Intelligence

Reasoning without cloud AI APIs.

[02]

Multi-Agent Automation

Complex tasks distributed across specialized agents.

[03]

Resilient Interaction

Accessibility-based automation supported by vision fallback.

[04]

Security & Privacy

Sanitized inputs, restricted execution, biometric authorization and encrypted local data.

[THE CORE IDEA]

"AI that can act on your computer,
— while the computer remains under your control."

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[AIR-GAPPED · PRIVATE · LOCAL-FIRST]

THANK YOU.

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